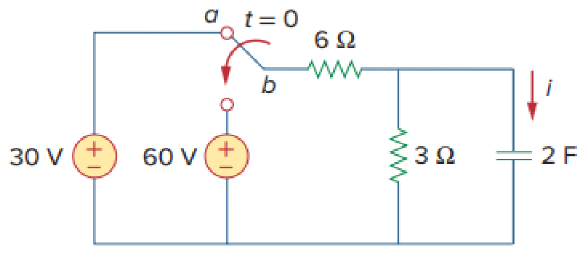
Chapter 7, Problem 44P

Problem

The switch in Fig. 7.111 has been in position *a* for a long time. At $t = 0$, it moves to position *b*. Calculate $i(t)$ for all $t > 0$.

Fig. 7.111:



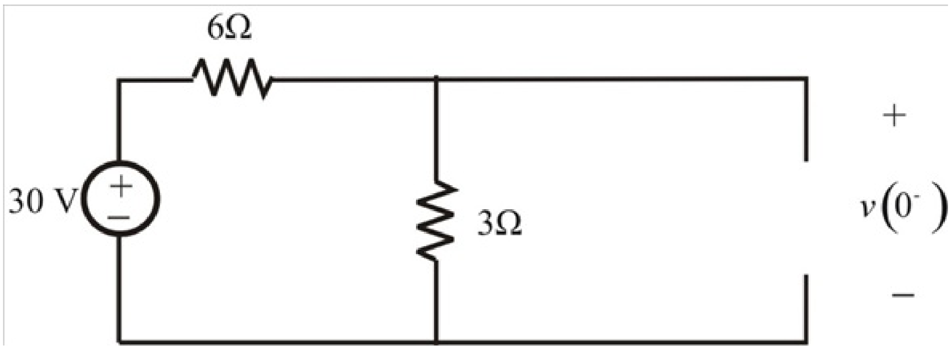
Step-by-step solution

Step 1 of 8

Refer to figure 7.111 in the textbook.

Consider that the switch is moved to position 'b' at $t = 0$, before that it was in position 'a' for a long time. The capacitor acts like an open circuit in steady state condition.

Draw the circuit for $t = 0^-$ as shown in the following figure:



Step 2 of 8

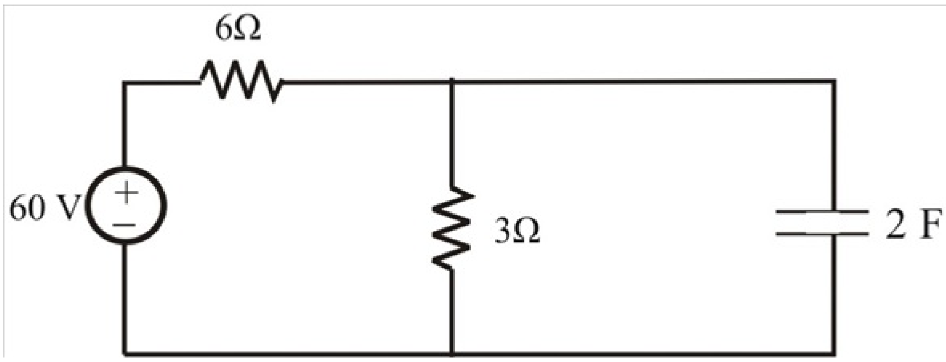
Apply the voltage division rule. Therefore, the initial voltage across the capacitor is,

$$v(0^-) = 30 \times \frac{3}{6+3}$$
$$v(0^-) = 10 \text{ V}$$

Capacitor doesn't allow sudden changes in voltage.

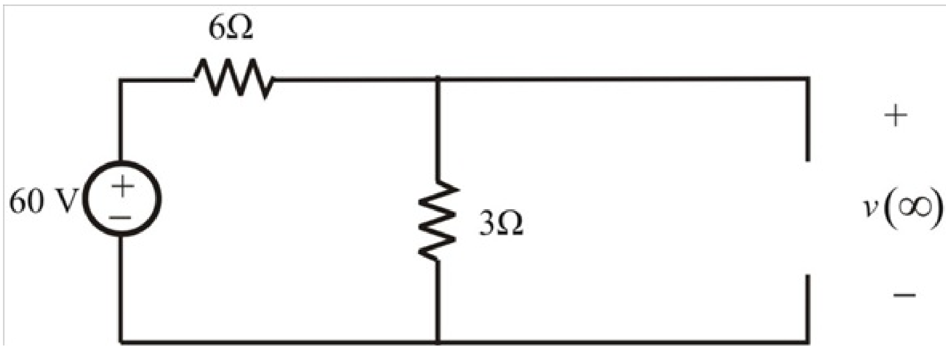
$$v(0^-) = v(0^+) = v(0) = 10 \text{ V}$$

Redraw the circuit for $t > 0$ as shown in the following figure:



Step 3 of 8

Switch is in that position for a long time the circuit reached steady state and capacitor acts like an open circuit again. We obtained $v(\infty)$ by using voltage division rule



Step 4 of 8

The steady state voltage across the capacitor is,

$$v(\infty) = 60 \times \frac{3}{6+3}$$
$$v(\infty) = 20 \text{ V}$$

The expression for the voltage across the capacitor is,

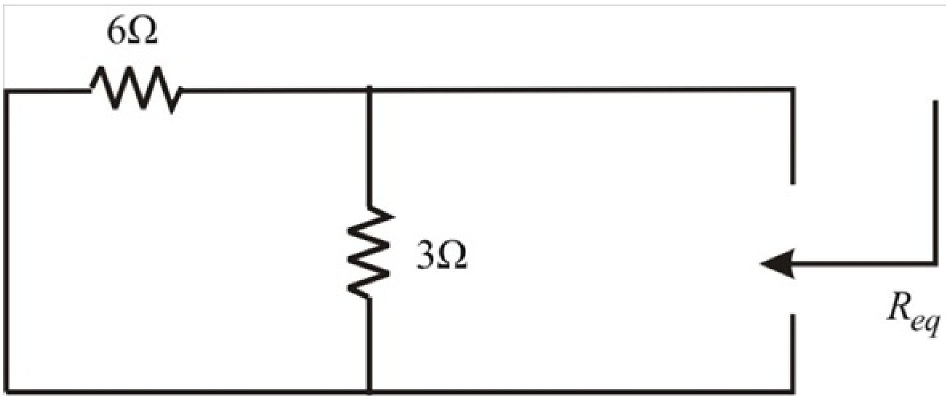
$$v(t) = v(\infty) + \left[v(0) - v(\infty)\right]e^{-\frac{t}{\tau}}$$
$$= 20 + \left[10 - 20\right]e^{-\frac{t}{\tau}}$$
$$v(t) = 20 - 10e^{-\frac{t}{\tau}} \text{ V}$$

Here, τ is time constant

$$\tau = R_{eq}C$$

Step 5 of 8

Determine the equivalent resistance R_{eq}. Consider the following circuit:



Step 6 of 8

Calculate the equivalent resistance R_{eq}.

$$R_{eq} = 3 \| 6$$
$$= \frac{3\times 6}{3+6}$$
$$R_{eq} = 2\ \Omega$$

Step 7 of 8

Calculate the time constant.

$$\tau = 2\times 2$$
$$\tau = 4\,\text{sec}$$

Step 8 of 8

Therefore, the voltage across the capacitor is,

$$v(t) = 20 - 10e^{-\frac{t}{4}}\ \text{V}$$
$$v(t) = 20 - 10e^{-0.25t}\ \text{V}$$

Therefore, the capacitor current is,

$$i(t) = C\frac{dv(t)}{dt}$$
$$= 2\frac{d}{dt}\left(20 - 10e^{-0.25t}\right)$$
$$= 2(-10)\left(-0.25e^{-0.25t}\right)$$
$$i(t) = 5e^{-0.25t}\ \text{A}$$

Therefore, the value of current $i(t)$ for all $t > 0$ is $\boxed{5e^{-0.25t}\ \text{A}}$.